structured lab exercises (Medium → Complex level) on **JavaScript Events**.

These are designed for:

- Practical classroom implementation
- Real-time DOM interaction
- Interview preparation
- Event-driven thinking

No solutions included — these are **student tasks**.

🧪 PART 1: Medium Level Exercises

🔹 Exercise 1 – Live Character Counter with Limit

Concepts Covered:

- input event
- keyup event
- preventDefault()
- Conditional logic

🎯 **Task:**

Create a textarea that:

- Shows live character count
- Maximum limit = 100 characters
- When limit reached:
    - Change border color to red
    - Disable submit button
- Show warning message

🔹 Exercise 2 – Dynamic List with Event Delegation

Concepts Covered:

- click event
- event.target
- event delegation
- DOM manipulation

🎯 **Task:**

Create a dynamic shopping list:

- Add items via input field
- Click item → mark as completed (toggle class)
- Click delete icon → remove item
- Only ONE event listener allowed for the list

🔹 **Exercise 3 – Keyboard Shortcut System**

Concepts Covered:

- keydown
- event.key
- Conditional logic

🎯 **Task:**

Implement:

- Press B → Change background color
- Press R → Reset page
- Press H → Show help alert
- Display pressed key on screen

🔹 **Exercise 4 – Image Hover Gallery**

Concepts Covered:

- mouseover
- mouseout
- click

🎯 **Task:**

Create 4 images:

- Hover → show image title
- Click → enlarge image
- Mouse out → restore original size

🔹 **Exercise 5 – Scroll Progress Indicator**

Concepts Covered:

- scroll event
- window properties
- dynamic style update

🎯 **Task:**

- Create long webpage
- Show scroll progress bar at top
- Change color when 80% scrolled

🧪 **PART 2: Upper-Medium Exercises**

🔹 **Exercise 6 – Smart Form Validation System**

Concepts Covered:

- focus
- blur
- input

- submit
- preventDefault()

🎯 **Task:**

Create a form with:

- Name
- Email
- Password

Requirements:

- Highlight input on focus
- Validate on blur
- Show error messages dynamically
- Prevent submission if invalid
- Show success message if valid

🔹 **Exercise 7 – Nested Event Propagation Tester**

Concepts Covered:

- bubbling
- capturing
- stopPropagation()

🎯 **Task:**

Create:

- Grandparent div
- Parent div
- Child button

Display order of event execution.
Add toggle button to:

- Switch between bubbling and capturing
- Enable/disable stopPropagation

🔹 Exercise 8 – Live Search Filter

Concepts Covered:

- keyup
- input
- Array filtering
- DOM update

🎯 **Task:**

Given a list of 20 names:

- Filter list in real-time
- Highlight matching letters
- Show "No Results" if empty

🧪 PART 3: Complex Level Exercises

🔥 Exercise 9 – Real-Time Dashboard (Multi-Event System)

Concepts Covered:

- click
- input
- resize
- scroll
- load
- event delegation

🎯 **Task:**

Build a dashboard that:

- Adds dynamic cards
- Deletes cards
- Tracks window size
- Shows scroll percentage
- Logs all events in activity panel

Constraint:

- Use event delegation for dynamic elements

🔥 **Exercise 10 – Custom Event System**

Concepts Covered:

- CustomEvent
- dispatchEvent
- addEventListener

🎯 **Task:**

Create:

- Button → "Generate Report"
- Dispatch custom event called reportGenerated
- Another component listens for it
- Display report summary on screen

🔥 **Exercise 11 – Multi-Step Form Wizard**

Concepts Covered:

- click
- submit
- input

- preventDefault
- conditional rendering

🎯 **Task:**

Create:

- Step 1 → Personal Info
- Step 2 → Address
- Step 3 → Confirmation

Requirements:

- Next button validation
- Back button support
- Show progress bar
- Final submit confirmation

🔥 **Exercise 12 – Advanced Event Delegation Grid**

Concepts Covered:

- delegation
- classList manipulation
- dataset
- event.target vs currentTarget

🎯 **Task:**

Create 5x5 grid of boxes:

- Click → toggle active
- Double click → change color
- Right click → disable box
- Show coordinates using dataset attributes

Only one event listener allowed for grid.

🧪 PART 4: Challenge (Interview Level)

🔥 Exercise 13 – Build a Mini Trello Board

Concepts Covered:

- Event delegation
- Multiple event types
- DOM creation
- Conditional logic
- Bubbling

🎯 **Task:**

Create:

- Columns (To Do, In Progress, Done)
- Add cards dynamically
- Click card → move to next column
- Double click → delete card
- Track total cards per column

Constraints:

- No inline events
- Use addEventListener only
- Use delegation